NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

ANTÓNIO MANUEL CARVALHO CANTEIRO

BSc in Computer and Electrotechnical Engineering

MISSILE INTERCEPTION WITH REACHABILITY TOOLS

DRAFT

Dissertation Plan
MASTER IN ELECTRICAL AND COMPUTER ENGINEERING
SPECIALIZATION IN CONTROL

NOVA University Lisbon
Draft: *January 30, 2026*

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Acronyms

6DoF Six Degrees of Freedom *(p. 10)*

CBF Control Barrier Function *(p. 6)*
CLF Control Lyapunov Function *(p. 6)*

GNC Guidance, Navigation and Control *(p. 3)*

HILS Hardware-in-the-Loop Simulation *(p. 2)*

LQR Linear-Quadratic Regulator *(p. 4)*

MILS Model-in-the-Loop Simulation *(p. 2)*
MPC Model-Based Predictive Control *(p. 5)*

NMPC Nonlinear Model-Based Predictive Control *(p. 5)*

PID Propotional-Integral-Derivative Control *(p. 4)*
PILS Processor-in-the-Loop Simulation *(p. 2)*
PN Proportional Navigation *(p. 3)*

QP Quadratic Programming *(p. 6)*

SILS Software-in-the-Loop Simulation *(p. 2)*

UAVs Unmanned Aerial Vehichles *(p. 1)*

Chapter 1

Introduction

Defending static structures against fast flying objects is an age-old problem that appears in a variety of scenarios, such as in aerospace (i.e. space debris endangering orbiting stations) and in military. It is in this latter scenario that the problem is more pressing and more challenging as the objects, in this case referred to as missiles, are both self-propelled and guided. A particular area of interest is active defense, where an unmanned autonomous vehicle is driven to intercept and neutralize the missile. Many technologies have been developed to this end, however these missile defense platforms often rely on specialized machinery that is either difficult to deploy *en-masse* or impractical in urban settings. Indeed, the twenty-first century has seen a rise in the need of missile defense in urban settings. At the same time there has also been a fast development of the capacities of commercial UAVs, which are inexpensive, lightweight, and easy to deploy when compared to traditional missile defense platforms. While much research has been done towards this end, resulting in a host of flight controllers that grant quad-rotor UAVs missile defense-capability, most methods do not provide early safety guarantees or trajectory feasibility analysis. A relatively novel method that may fill these gaps is reachability analysis.

Thus, the goal of this study is to design a flight controller for a quad-rotor UAV such that it defends a critical structure from a missile heading towards it, whether by intercepting it or deviating the missile's trajectory from the structure. The strategy considered for the controller design will be based on reachability tools, which utilises the concept of a reachable set.

1.1 State of the Art

The topics relevant to this project include that of modelling Unmanned Aerial Vehichles (UAVs), specifically of missiles and quad-rotor drones; control techniques for guided missiles, specifically with stationary targets, and with adversary avoidance; control techniques for quad-rotor drones, specifically with non-cooperative intercept targets. Therefore it's

necessary to map out the state of the art on these topics.

1.1.1 Simulation

It would be impractical to iterate the design of such a control function based on real flight tests alone, specially in the early stages of the project, as each change would require a new, lengthy flight test. Therefore it is necessary to simulate the scenario under study, meaning, to simulate the missile and drone dynamics such as the aerodynamic effects acting on the vehicles, in response to input actuations. An almost ubiquitous tool used in aerodynamical simulation and control design are MATLAB/Simulink [1], thanks to the diversity of applications and efficiency they provide. Other tools such as X-Plane and Gazebo-ROS provide a visual simulation environment with MATLAB integration. Study [2] details a systematic methodology of running simulation techniques that sequentially integrate more and more of the actual system into the simulation. These are Model-in-the-Loop Simulation (MILS), Software-in-the-Loop Simulation (SILS), Processor-in-the-Loop Simulation (PILS) and Hardware-in-the-Loop Simulation (HILS), near-actual simulation techniques that can help speed up the flight system's development.

1.1.2 Modelling

1.1.2.1 UAV Dynamics

Obtaining an accurate mathematical model for the vehicle dynamics is crucial, whether it be for simulation or designing the flight systems. Regarding the modelling of missile dynamics, the most relevant phases of flight for this study are the midflight and homing phases. The rotational dynamics of the missile may be parametrized with Euler angles or quaternions. When it comes to quad-rotor drone dynamics, this study is most concerned with the post-launch flight phase. In [3], it is shown that the 6DoF dynamic model of a quad-rotor are complicated, strongly nonlinear, inherently unstable with coupled system variables and under-actuated. However, they may be thought of as two distinct control problems; the outer loop tracking the position and the inner loop tracking the attitude, which should simplify the controller design process.

1.1.2.2 Aerodynamic Effects

Both the missile and the quad-rotor are expected to move at very high speeds. Therefore, it's clear that aerodynamic effects such as air drag must be considered in the modelling process. In the case of quad-rotors, blade flapping and thrust variations pose a significant disturbance to the system, and must also be accounted for. In [4], these effects are explicitly compensated for in the control loop of a quad-rotor drone.

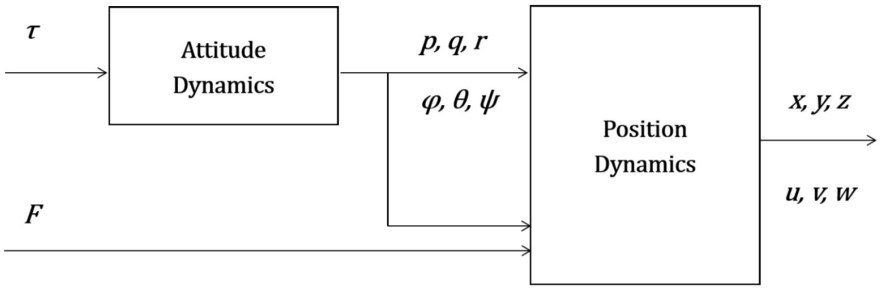


Figure 1.1: Diagram of UAV model. Source: [3]

1.1.3 Control Systems

It's worthwhile to review existing flight control systems, for both missiles and drones. The most widely used methodologies for missile control systems are Proportional Navigation (PN) guidance, and for UAVs they are Optimal Control-based. These may further be classified into Linear Control Systems and Model Based Nonlinear Control Systems [5].

1.1.3.1 GNC

The systems that control the movements of UAVs can be described by their three core functions: Guidance, Navigation and Control (GNC). Guidance refers to the systems that determine the desired paths from the vehicle's current position to the target waypoint, as well as establishing the changes in velocity, rotation or acceleration to achieve this course. Navigation refers to the systems that determine, at a given instant of time, the vehicle's location, velocity, rotation, and angular rate. These variables are the system's state vector. Control is referring to the manipulation of forces and torques, by means of actuators (i.e., steering device, thrusters, etc.), needed to follow the desired path while maintaining vehicle stability [6]. The goal of this project is, essentially, to design a control function that given the current state vector of both the missile and the quad-rotor drone, will drive the drone to intercept the missile.

1.1.3.2 Linear Control Methods

Linear control methods are the least computationally complex methods and the easiest to implement, and generally consist in driving the controlled system variable to converge to a desired setpoint, based on negative feedback. However, they tend to have limited robustness when the desired value is variable, or when the vehicle is under-acted, this is, when there are fewer actuators than the number of degrees of freedom to be controlled. To overcome these limitations, the controlled system must be linearized, usually by finding linear dynamics around a stable operating point or trajectory.

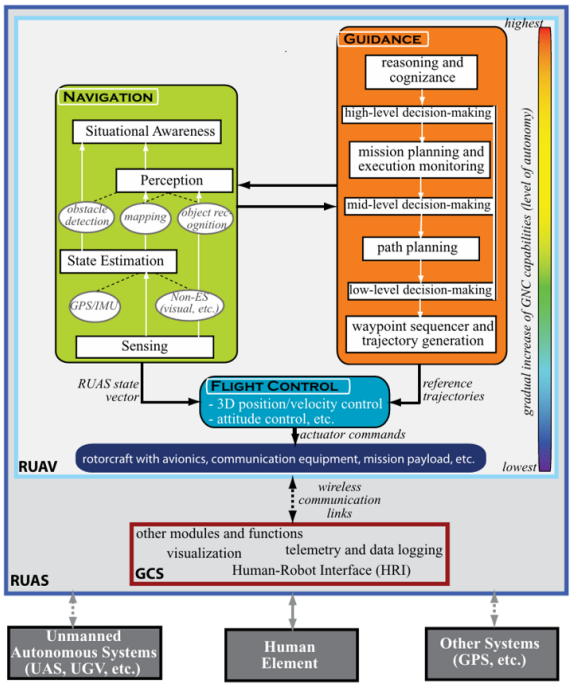


Figure 1.2: The overall structure of guidance, navigation, and control systems onboard a RUAS. Source: [5], Figure 3

PID One of the most common Linear control methods is the Propotional-Integral-Derivative Control (PID) control. PID is based on negative feedback, and it aims to converge the controlled variable's value to a given setpoint, according to given specifications. From these specifications the proportional, integral, and derivative gain are selected. These will be multiplied by the respective parts of the error signal, meaning the difference between the desired setpoint and the current value of the controlled variable. This transformed signal is injected back into the system as input. In [7] a survey of the performance of different PID methods on UAVs is conducted. It reveals that linear PIDs are effective at hovering, moving to a waypoint and following a trajectory, with the advantages of being simple to design and implement as well as not needing knowledge of the system model. It has several disadvantages however, such as requiring the model to be linearized, it is not robust to external disturbances and it's only valid only a specific operation point under specific conditions.

LQR Another commonly used linear control method is the Linear-Quadratic Regulator (LQR). The system dynamics are described by a set of linear differential equations and the cost is described by a quadratic function, usually defined as the geometric norm of the difference between the state vector and the setpoint at the current instant. Weighting factors are selected then applied to different parts of the cost function, such as the final value or the sum of all deviations. The control actuation may also feature in the cost function. The LQR provides the optimal control law given these parameters and is robust to perturbations. Moreover, it can handle time-varying models. In [8], a quad-rotor drone

is controlled with an LQR based controller. It is shown that this controller is robust, with low steady state error.

Gain Scheduling Gain Scheduling is another linear control method, in which non-linear system dynamics are modelled as a collection of simplified linear models, with each model representing a particular operating regime. Each model has a corresponding linear control law. In [9], a gain scheduling-based controller is implemented in a quad-rotor drone, where different PID controllers are used depending on the current operating regime. The study shows that, relative to the singular PID controller, the performance in trajectory tracking is increased, while maintaining the advantages of PIDs.

1.1.3.3 Nonlinear Control Methods

Model-Based Nonlinear Controllers are another family of flight control systems. They have the advantage of being able to handle systems' nonlinear dynamics, making them more robust and versatile than linear control methods. This comes at the cost of a more complex, more computationally expensive implementation.

MPC Model-Based Predictive Control (MPC) as a model-based control method is an extension of the LQR. However, the MPC estimates the model output in a finite time horizon then minimizes tracking error for this horizon. Usually, the control law is updated online. Designing the MPC controller requires an explicit linear model of the system, and this is not usually the case in UAV control, making linearization necessary. Study [10] considers both a UAV dynamic target interception scenario and a surveillance scenario, and proposes a novel MPC design with variable time steps. The controller was tested both in simulation and real-world flight tests, in 2D and 3D space, and was shown to be effective and efficient.

NMPC Nonlinear Model-Based Predictive Control (NMPC) is a variation of the linear MPC that can handle nonlinear models. The optimization problem is usually non-convex, therefore this method does not usually yield a closed-form expression and requires numerical methods to obtain the optimal control law for this time step. This introduces the "curse of dimensionality", where the complexity grows exponentially with the horizon. Study [11] tackles the problem of collision avoidance in a multi-UAV agricultural scenario. It compares the performance of three NMPC-based control strategies: distributed NMPC where the NMPCs of the UAVs are coupled through nonlinear constraints, distributed NMPC where the NMPCs of the UAVs are coupled through penalty costs, and imposing flight corridors by modelling uncertainties and disturbances. It shows that in a scenario with no path intersections the flight corridor strategy ranks better, however it fails to ensure collision avoidance when there are path intersections, while the other two methods are generally effective but with varying performances.

Backstepping Backstepping Control is a nonlinear control method, commonly used in systems with a relative degree higher than one. This method separates the system into various subsystems such that the relative degree of each is one, then a "virtual control law" is applied to the higher-order subsystem such that it stabilizes the lower order one. This creates a series of "virtual controllers" that ultimately allow the controlled variables to converge towards the setpoint. This allows for an iterative, systematic design process that especially favors the layered structure of UAV dynamics. However, the complexity of these controllers rises dramatically as the relative degree increases, since each differentiation step is increasingly complicated to solve analytically. Study [12] compares the performances of two route-planning approaches for a UAV controller with collision avoidance based on backstepping: a safety ball approach where the UAV stays a minimum distance away from obstacle boundaries while converging towards the target, and a point-mass approach that treats obstacles as a sphere with radius equal to the obstacle's largest dimension. It shows that the safety ball approach has better performance when it comes to computational complexity, but does not minimise the tracking error.

CLF-CBF Control Lyapunov Function (CLF), along with Control Barrier Function (CBF) and unified by Quadratic Programming (QP) establish their own class of nonlinear controllers. The CLF uses the Lyapunov function, which represents an objective. With the CLF, the input norm is minimized while subject to an inequality constraint on the derivative of a Lyapunov function, and the system is guaranteed to converge exponentially towards the objective. CLF methodology allows the system to have several objectives simply by adding CLFs representing them to the constraints. CBFs are similar to CLFs, except they use Lyapunov-like functions called Barrier functions to define a boundary that the system cannot cross, and will diverge away from [13]. By joining CLFs and CBFs, a powerful, a robust controller is achieved. Other methods may be used along with CLF/CBF controllers to further expand their efficacy, such as backstepping. The study [14] proposes a defense-winning strategy for active target protection involving multiple nonlinear players, utilising a CBF-QP controller.

1.1.3.4 Reachability-Based Guidance

PN and optimal control based methods, while traditionally used have their limitations. PN guidance cannot account for changes in the adversary's behavior, while optimal control cannot guarantee feasibility given the current conditions. A modern alternative is Reachability based guidance, that aims to grapple with these limitations. Reachability analysis can be used to implicitly guarantee feasibility of a trajectory, or for collision avoidance. Study [15] employs a novel method for determining the reachable set boundaries for aerodynamic interceptors.

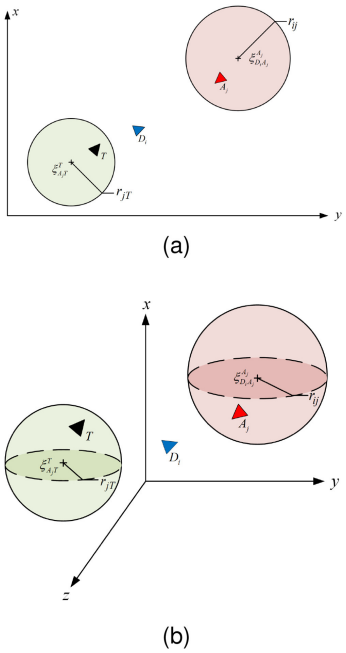


Figure 1.3: Illustration of the m-sphere enclosure region. (a) 2-sphere enclosure region in a 2D plane. (b) 3-sphere enclosure region in the 3D space. Source: [14], Figure 1

Approximate Geometric Method One way to calculate the reachable set is the Approximate Geometric Method. It approximates the reachable set of the model by manipulating convex shapes such as ellipsoids, zonotopes, or polytopes such that they closely fit the bounds of the expected set in the next time steps. This method can even be used with nonlinear models, as shown in [16]. A drawback of these methods is that, as the set computation is iterated for each following time step, the dimensional complexity grows exponentially, thus increasing the computational burden of this method. However, study [17] proposes an algorithm that significantly reduces this effect.

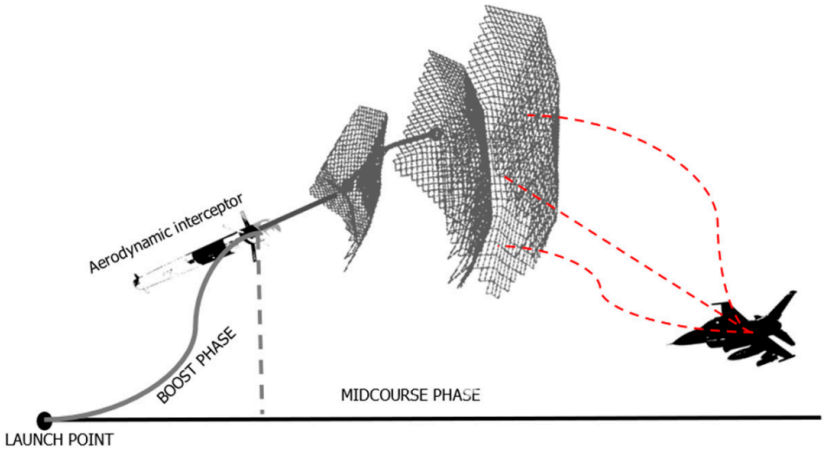


Figure 1.4: Example attainable zone. Source: [17], Figure 2

Chapter 2

Problem Formulation

Notation The set of real numbers is denoted by $\mathbb{R}$. Given a vector $\vec{v} \in \mathbb{R}^n$, its transpose is denoted by $\vec{v}'$, and likewise for a matrix $\mathbf{M}$. The operation diag($\vec{v}$) creates a matrix with diagonal elements v and off-diagonal elements 0. The inner product between two matrices or vectors is denoted by $\vec{v} \cdot \mathbf{M}$. The cross product operation between a vector and a matrix, denoted by $\vec{v} \times \mathbf{M}$, is equivalent to the inner product between the skew asymmetric matrix of that vector and that matrix, which can be denoted by $[\vec{v}] \cdot \mathbf{M}$. The operation vec($\mathbf{M}$) returns the vectorized matrix.

When expressing the coordinates of a point or vector on an object, it may be expressed in the inertial reference frame, denoted by $\mathcal{F}_I = (\vec{e}_I, \vec{e}_J, \vec{e}_K)$, which refers to the coordinate system whose origin is fixed on the ground, or the body reference frame, denoted by $\mathcal{F}_B = (\vec{e}_x, \vec{e}_y, \vec{e}_z)$, which has it's origin on the center mass of the object. When a vector has an uppercase B in overscript ($\vec{v}^B$), it's expressed in the body reference frame. In the absence of an overscript notation, assume it's expressed in the inertial reference frame.

2.1 Modeling

Designing the quad-rotor's flight controller will require both vehicle's mathematical models that capture their flight dynamics as accurately as possible.

2.1.1 Missile

The missile's model is built on the assumptions that it's in the midflight guidance phase, mass steadily decreases as fuel is spent, method of propulsion is a two-pulse rocket motor that provides constant downward acceleration (in $\mathcal{F}_B$), the attitude actuators are a ring of pulse jets near the top, it has GNC capability that allows it to discern its position relative to obstacles and the target and path plan, and it is subject to aerodynamic drag.

The missile's model is represented by the state vector $x_R = [\vec{p}_R, \vec{v}_R, \vec{\omega}^B, \text{vec}(\mathbf{A})]$, considered to be a point-mass at the missile's center of mass, in which

$\vec{p}_R \in \mathbb{R}^3$ is the position of the center of mass, expressed in $\mathcal{F}_I$,

$\vec{v}_R \in \mathbb{R}^3$ is the linear speed of the center of mass, expressed in $\mathcal{F}_I$,

$\vec{\omega}^B \in \mathbb{R}^3$ is the angular velocity of the rocket, expressed in $\mathcal{F}_B$, and

$\mathbf{A} \in \mathbb{R}^{3\times3}$ is Euler's rotational matrix of the rocket, expressed in $\mathcal{F}_I$.

Additional variables are defined as

$\vec{a}^B \in \mathbb{R}^3$ is the thruster acceleration vector acting on the rocket, expressed in $\mathcal{F}_B$,

$\vec{g} \in \mathbb{R}^3$ is the gravitational acceleration vector, expressed in $\mathcal{F}_I$,

$\mathbf{J} \in \mathbb{R}^{3\times3}$ is the rocket's inertia matrix, and

$\vec{\tau}^B \in \mathbb{R}^3$ is the torque vector on the rocket body, expressed in $\mathcal{F}_B$.

Furthermore, $\mathbf{J}$, $\vec{\tau}^B$ and $\vec{a}^B$ are given by the expressions

$$\vec{\tau}^B = \begin{bmatrix} 0 \\ 0 \\ -L \end{bmatrix} \times (m \cdot \vec{a}^B),$$

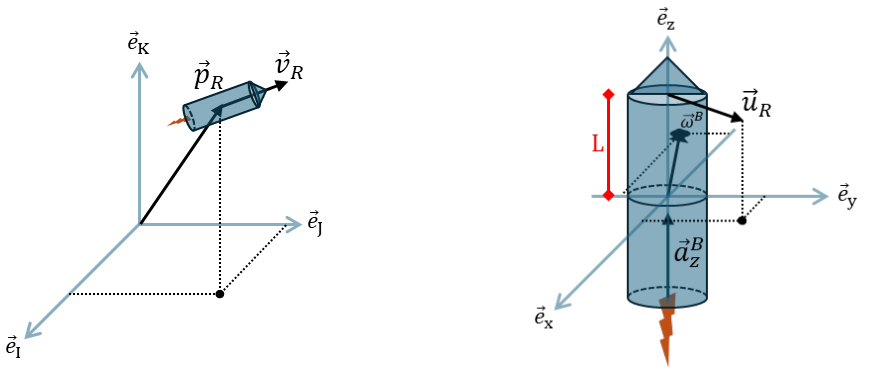
$$\mathbf{J} = \text{diag}(\frac{1}{4}m \cdot R^2 + \frac{1}{12}m(2L)^2, \quad \frac{1}{4}m \cdot R^2 + \frac{1}{12}m(2L)^2, \quad \frac{1}{2}m \cdot R^2),$$

$$\vec{a}^B = \begin{bmatrix} a_x^B \\ a_y^B \\ a_z^B \end{bmatrix}.$$

The lateral acceleration input vector $u_R \in \mathbb{R}^2$ is defined as

$$u_R = \begin{bmatrix} a_x^B \\ a_y^B \end{bmatrix}$$

Figure 2.1 shows the missile's state variables, apart from $\mathbf{A}$, in their respective referential. Even though, mathematically speaking, each vector actually originates from the origin of their referential, here they have been represented as originating from their "point of application" for a more intuitive viewing.



(a) Missile's state variables in $\mathcal{F}_I$ (b) Missile's state variables in $\mathcal{F}_B$

Figure 2.1: Missile's state vector diagram

The variable's dynamics are thus defined by the equations

$$\begin{aligned}
\dot{\vec{p}}_R &= \vec{v}_R = f_p(x_R), \\
\dot{\vec{v}}_R &= \mathbf{A} \cdot \vec{a}^B + \vec{g} = f_v(x_R, u_R), \\
\dot{\vec{\omega}}^B &= \mathbf{J}^{-1} \cdot (\vec{\tau}^B - \vec{\omega}^B \times \mathbf{J} \cdot \vec{\omega}^B) = f_\omega(x_R, u_R), \\
\mathrm{vec}(\dot{\mathbf{A}}) &= \mathrm{vec}(-\vec{\omega}^B \times \mathbf{A}) = f_A(x_R).
\end{aligned}$$

The missile's differential dynamics are defined as $\dot{x}_R = f_R(x_R, u_R)$, and

$$f_R(x_R, u_R) = \begin{bmatrix} f_p(x_R) \\ f_v(x_R, u_R) \\ f_\omega(x_R, u_R) \\ f_A(x_R) \end{bmatrix}.$$

2.1.2 Quad-Rotor Drone

The quad-rotor drone's Six Degrees of Freedom (6DoF) model is built on the assumptions that its mass remains constant, the method of propulsion are four vertical rotors, the attitude actuators are servos attached to the rotors that provide thrust vectoring, it has GNC capability that allows it to discern its position relative to obstacles and the target and path plan, and it is subject to aerodynamic drag.

The drone's model is represented by the state vector x_D, considered to be a point-mass at the drone's center of mass.

The drone's differential dynamics are defined as $\dot{x}_D = f_D(x_D, u_D)$.

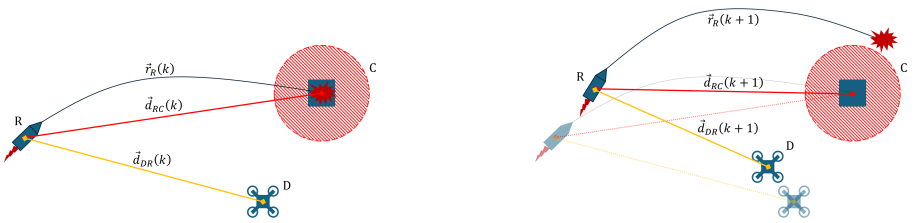
2.2 Missile Defense

The scenario in study can be thought of a defense-intercept game with two asymmetric players, with each player having their hierarchy of objectives to satisfy.

The missile's hierarchy of objectives are static target interception and dynamic pursuer evasion. The drone's hierarchy of objectives are dynamic target interception and static area defense.

Each objective poses its own control problem, but each player's GNC structure must account for all its objectives. Regarding a player's navigation system, it must be able to estimate its own and the opponent's state relative to each other and relative to the critical area's position. As for the guidance system, it must apply decision-making based on the player's navigation data and it's goals to plan the most optimal trajectory. Finally, the flight control system calculates the appropriate actuation commands given the player's dynamics, based on its navigation and guidance data.

Figure 2.2 illustrates both the missile's and drone's expected behaviors in a successful defense scenario. C identifies the critical structure, D identifies the drone, R identifies the missile, $\vec{r}_R$ is the missile's best feasible planned trajectory, $\vec{d}_{DR}$ is the distance vector between the missile and the drone, and $\vec{d}_{RC}$ is the distance vector between the missile and the critical structure. From instant k to instant $k+1$ the distance vectors shrink, and the missile's planned trajectory can no longer hit the target thanks to its collision avoidance.



(a) At instant k (b) At instant k+1

Figure 2.2: Missile defense evolution in time

For the flight control system, it is necessary to define a control law. Since either player is described by a non-linear model, several methods may be used, such as model linearization around an ideal trajectory or NMPC. Optimal Control-based methods, which obtain the optimal input actuation by minimizing some cost function related to the objective, stand out for their ability to join guidance with control, as they do not need explicit path planning and trajectory generation to generate the optimal input. These characteristics favor the control problems at hand, seeing that players do not need to follow specific trajectories, only to converge on their targets while satisfying their other objectives. CLF/CBF-type

controllers are also well suited to these problems, since the evasion and defense objectives are essentialy barrier conditions.

For the scenario in which the rocket does not evade the drone, let $V_R : \mathcal{X} \to \mathbb{R}$ a quadratic function representing the distance between the rocket and the critical structure. The rocket's objective can be represented by the optimization problem:

$$\begin{aligned} \min_{u} \quad & V_R(x, u) \\ \text{s.t.:} \quad & \dot{x} = f_R(x, u) \\ & x \in \mathcal{X}_R \\ & u \in \mathcal{U}_R \end{aligned}$$

The drone's objective is dual; to intercept the rocket as well as diverting the rocket away from the critical structure. Let $V_D : \mathcal{X} \to \mathbb{R}$ a quadratic function representing the distance between the drone and the rocket, and $V_{\text{crit}} : \mathcal{Z} \to \mathbb{R}$ a quadratic CBF representing the intersection between the reachable set of the rocket and the critical area. Furthermore, consider $\gamma(\cdot)$ is a $\mathcal{K}$ class function. The drone's objective can be represented by the optimization problem:

$$\begin{aligned} \min_{u} \quad & V_D(x, z, u) \\ \text{s.t.:} \quad & \dot{x} = f_D(x, u) \\ & \dot{V}_{\text{crit}}(z) \leq -\gamma(V_{crit}(z)) \\ & x \in \mathcal{X}_D \\ & z \in \mathcal{X}_R \\ & u \in \mathcal{U}_D \end{aligned}$$

Chapter 3

Project Planning

3.1 Tasks

The project can be split into tasks, described bellow.

1. **Literature Review:** A comprehensive survey on the literature of the relevant topics is conducted, with the goal of building the theoretical foundations necessary for practical implementation. This literature review will address the topics of designing UAV controllers and reachability analysis, as well as any and all sub-topics branching out from these.

2. **Definition of Requirements:** The problem will be explicitly formulated. The specific requirements of the "success" conditions of the study will be identified.

3. **Open-Loop Design:** This phase consists of defining the mathematical model of both the drone and the missile dynamics. These models must be accurate, considering the full physical dynamics, and complete, considering aerodynamic effects.

4. **Defense-Intercept Algorithm Definition:** The reachability set-based control function is theoretically defined such that it meets the defense-intercept requirements previously laid out. The control function is implemented in the closed-loop of the drone model.

5. **Controller Validation in Simulation:** The drone controller is systematically validated according to the MILS-SILS-PILS-HILS techniques. The simulation scenarios must also be defined such that they broach as wide as possible a range of real scenarios, by considering more or less sophisticated missile controllers and by considering adverse flight conditions.

6. **Test Flight Validation:** The control function is loaded into the physical hardware of the drone and flight tests are conducted to validate its effectiveness and evaluate its performance. In these tests, the drone is facing off against a fixed-wing UAV that mimics a missile.

7. **Writing Thesis:** The final document is written, describing the work done, evaluating the results and drawing up conclusions.

3.2 Planned Schedule

Table 3.1 outlines the planned distribution of tasks over the time available, in the form of a Gantt Chart.

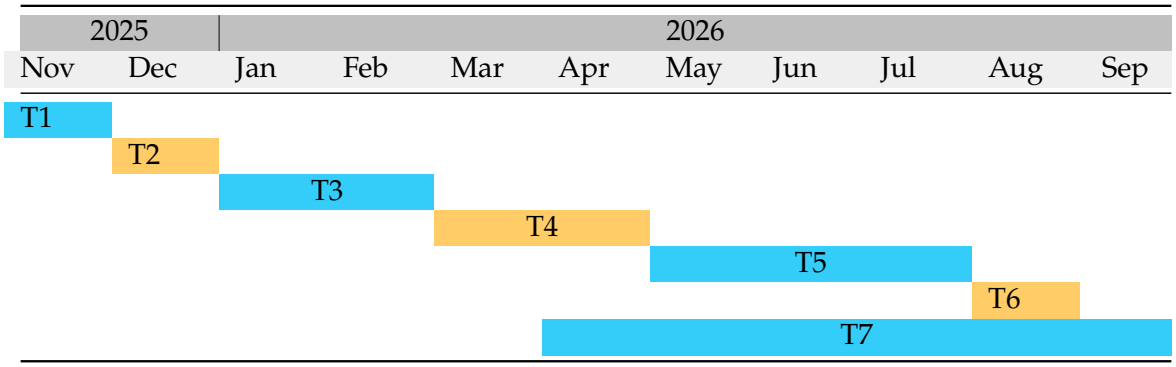


Table 3.1: Gantt Chart

3.3 Prior Work

In preparation for this study, the continuous mathematical model for the missile dynamics was derived, ignoring aerodynamical effects. This model is described in Subsection 2.1.1. This open-loop model was implemented in MATLAB, as well as the framework for simulations. The framework was made with a closed-loop structure in mind, running in discrete time with a sampling period of 0.1 seconds, so that control inputs may be changed time-step to time-step. Despite this the model used in the simulation is continuous, as the MATLAB function ode45 is used to compute the values in the following time-step, and the sampling period is sufficiently small to ignore distortions.

Figure 3.1 shows a missile simulation meant to verify if the implemented dynamics behave as expected. The initial position of the base of the missile is at the origin, velocity is zero, angular velocity is zero and the rotation matrix is identity, so that the missile sits upright. The blue vector represents the missile body, magenta are the thrust acceleration vectors acting on the missile, and red is the trajectory of the missile until this instant.

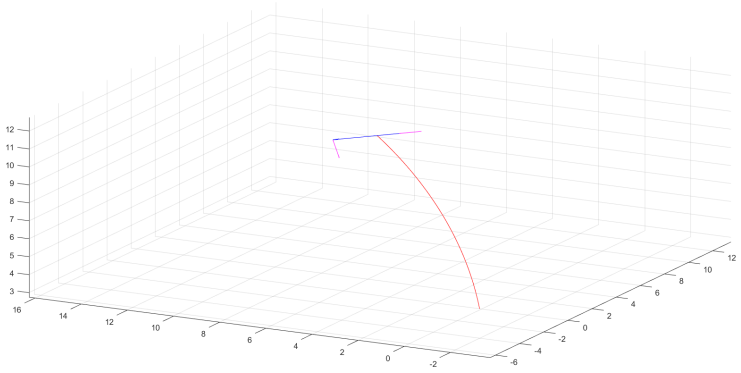


Figure 3.1: Simulation of previous work.

Chapter 4

Conclusions

This project preparation report has introduced the problem to be studied and the motivations behind it. In Section 1.1 it has presented a comprehensive review on the state of the art on modelling and simulating UAVs, as well as the kind of UAV flight controllers used in interception and defense scenarios, and on reachability analysis. Finally, in Chapter 3 it outlined the tasks the project will consist of, the planned schedule and prior work done.

When it comes to the UAV interception problem, the literature reveals that there is a gap in research of reachability analysis-based controllers. This study aims to address that gap, and design an effective and efficient reachability analysis-based controller with practical applications, such that it can provide a UAV with missile defense capabilities with early success guarantees, as well as other advantages granted by reachability analysis.

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